

Project 7: NAT Gateway – Private Subnet Internet Access

Objective

The objective of this project is to provide outbound internet access to EC2 instances residing in a private subnet without assigning them a public IP address. This configuration allows private instances to safely download software updates, install security patches, and access external web services while remaining completely shielded and inaccessible from the public internet.

Step 1: Launch Bastion Host (Public EC2)

The Bastion Host acts as a secure jump box to access the EC2 instance located inside the private subnet.

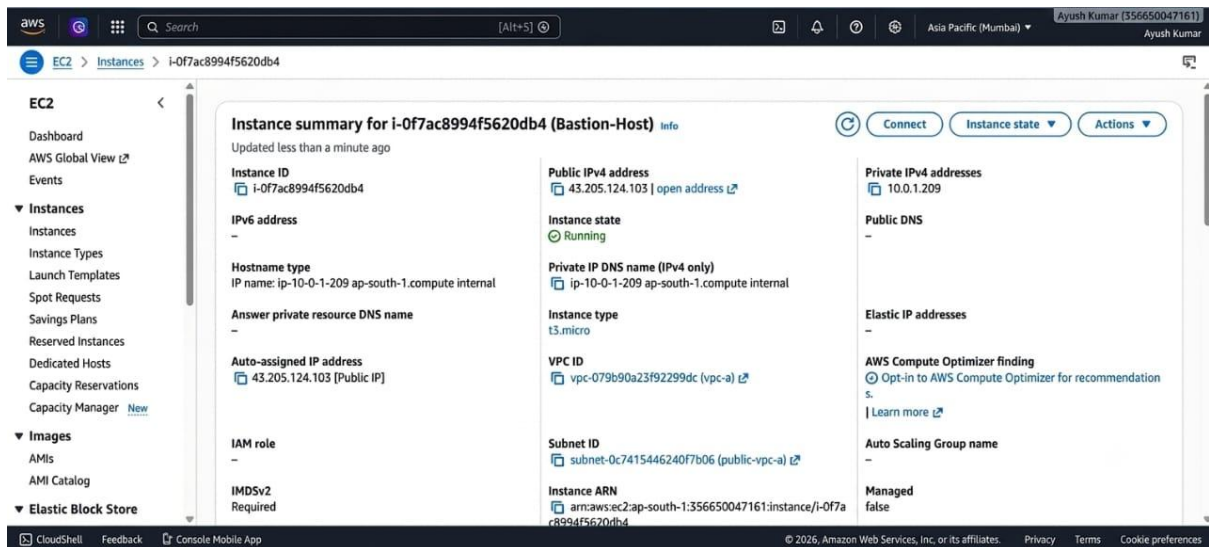
Instance Configuration

Setting	Value
Name	Bastion-Host
AMI	Amazon Linux 2023
Instance Type	t2.micro
VPC	Project-VPC
Subnet	Public-Subnet
Auto Assign Public IP	Enable
Security Group	EC2-SG

Security Group Inbound Rules

Type	Port	Source
SSH	22	My IP

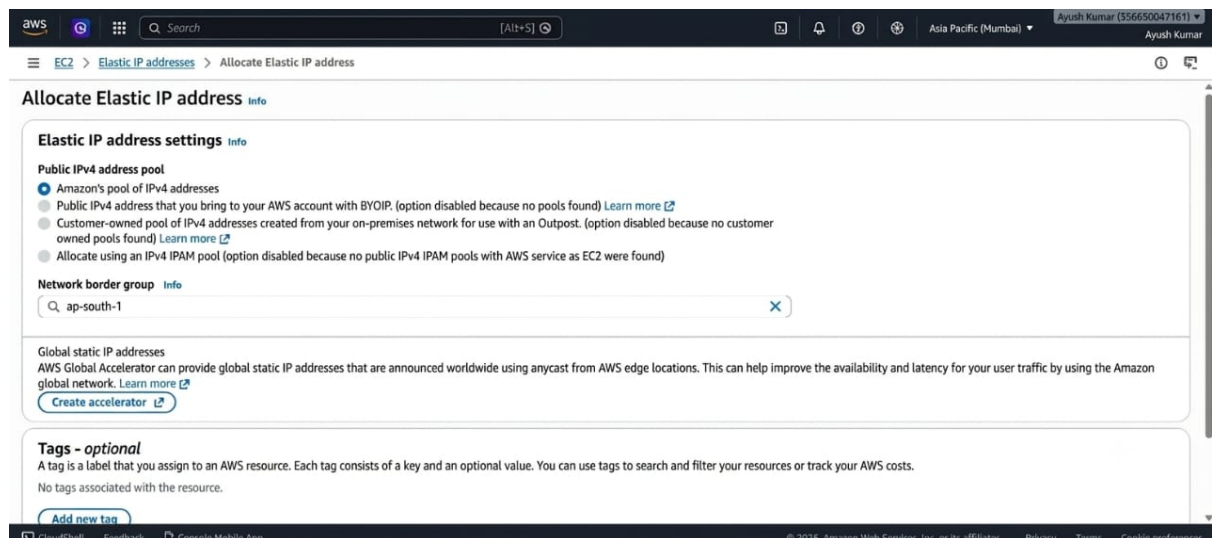
Action: Review and click **Launch Instance**.



Step 2: Allocate an Elastic IP

An Elastic IP is required so that the NAT Gateway has a static public IP address to communicate with the internet.

1. Open the **EC2 Console**.
2. Navigation pane me **Elastic IPs** par click karo.
3. **Allocate Elastic IP address** select karo.
4. Settings default rehne do aur **Allocate** par click karo.



Step 3: Create a NAT Gateway

The NAT Gateway must always be placed inside a public subnet so it can route traffic to the Internet Gateway.

Navigation: VPC Console → NAT Gateways → Create NAT Gateway

NAT Gateway Configuration

Setting	Value
Name	Project-NAT
Connectivity Type	Public
Subnet	Public-Subnet (<i>Important: Select Public Subnet</i>)
Elastic IP	Select the Elastic IP allocated in Step 2

Action: Click **Create NAT Gateway** and wait for the status to change to **Available**.

Create NAT gateway Info
A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
Project-NAT
The name can be up to 256 characters long.

Availability mode Info
Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☐ Regional - *new*
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☒ Zonal
Provides granular control within a specific availability zone, adhering to subnet level settings.

Subnet
Select a subnet in which to create the NAT gateway.
subnet-0c7415d46240f fb06 (public-vpc-a)

Connectivity type
Select a connectivity type for the NAT gateway.
☒ Public
☐ Private

Elastic IP allocation ID Info
Assign an Elastic IP address to the NAT gateway.

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Step 4: Update Private Route Table

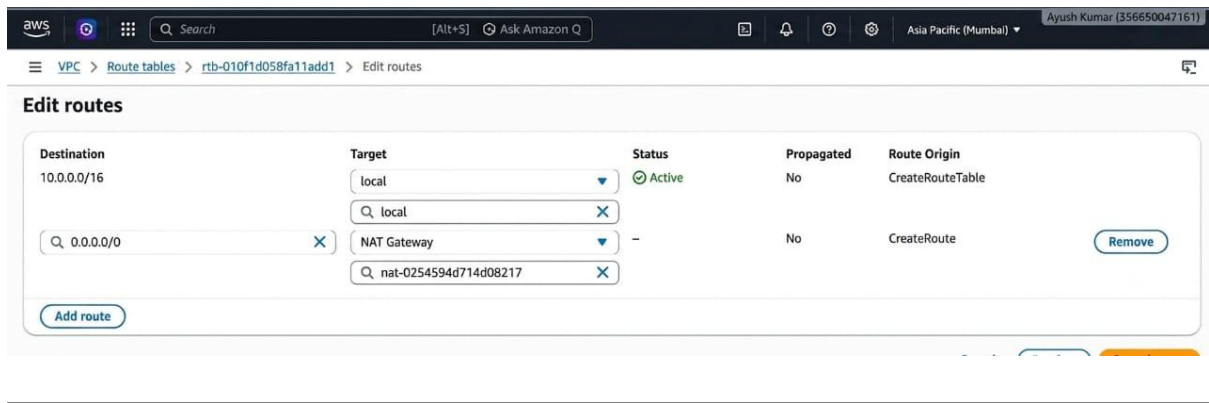
To route traffic from the private subnet to the internet, we must point the default route (0.0.0.0/0) to the NAT Gateway.

Navigation: **VPC Console** → **Route Tables** → Select **Private-RT** → Go to **Routes** tab → Click **Edit Routes**

Route Entry

- **Destination:** 0.0.0.0/0
- **Target:** NAT Gateway (Select your Project-NAT ID)

Action: Click **Save Changes**.



Step 5: Launch Private EC2 Instance

Now, create the secure backend server inside the private subnet without any public interface.

Instance Configuration

Setting	Value
Name	Private-Server
AMI	Amazon Linux 2023
Instance Type	t2.micro
VPC	Project-VPC
Subnet	Private-Subnet
Auto Assign Public IP	Disable
Security Group	EC2-SG (Allows inbound SSH from the Bastion Host)

Verification: Once launched, verify that the **Public IPv4** field for this instance is completely **None/Empty**.

Step 6: Connect to the Bastion Host

Open your local terminal/command prompt and log into your public-facing jump server using your private key.

Bash

Execute from your local machine

```
ssh -i ayush-kumar.pem ec2-user@<Bastion-Public-IP>
```

Example: `ssh -i ayush-kumar.pem ec2-user@43.205.124.103`

Step 7: Connect from Bastion Host to Private EC2

Copy your private key file over to the Bastion Host to jump securely into your private instance.

Bash

Step 7a: Securely copy key from local machine to Bastion Host

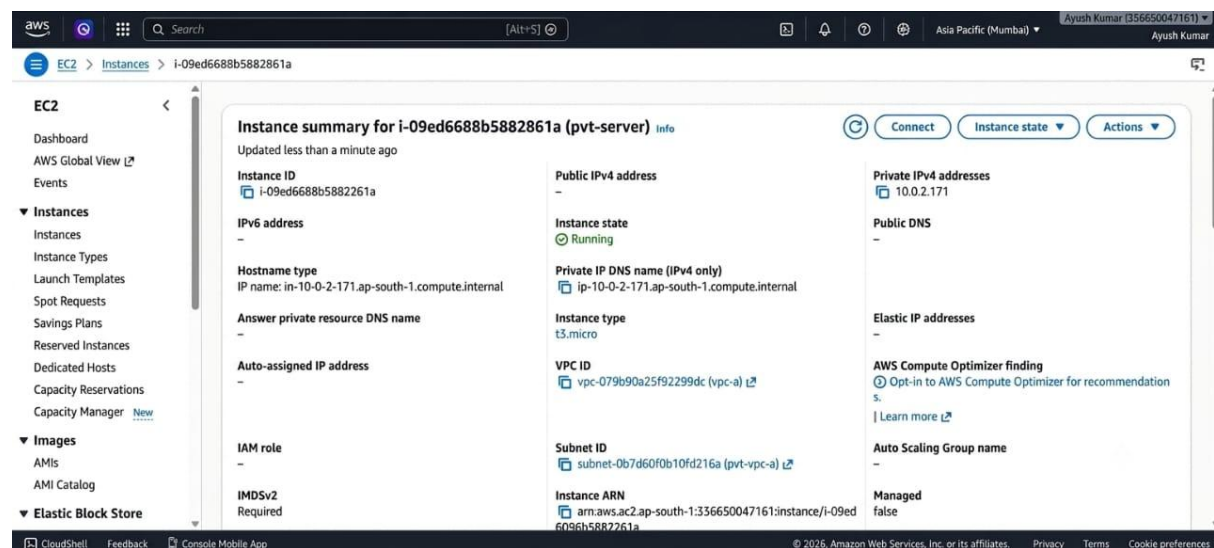
```
scp -i ayush-kumar.pem ayush-kumar.pem ec2-user@<Bastion-Public-IP>:/home/ec2-user/
```

Step 7b: Inside Bastion Host, modify key permissions

```
chmod 400 ayush-kumar.pem
```

Step 7c: Connect to Private EC2 using its Private IP address

```
ssh -i ayush-kumar.pem ec2-user@<Private-EC2-Private-IP>
```



Step 8: Test Outbound Internet Access

Run the following commands inside the **Private EC2** shell to confirm that the NAT Gateway is correctly translating traffic:

Bash

1. Test package manager updates

```
sudo dnf update -y
```

```
curl -I https://www.google.com
```

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\AYUSH KUMAR> cd Downloads
PS C:\Users\AYUSH KUMAR\Downloads> scp -i ".\ayush.pem" ".\ayush.pem" ec2-user@43.205.124.103:/home/ec2-user/ayush.pem
100% 1678   23.1KB/s   00:00
PS C:\Users\AYUSH KUMAR\Downloads> ssh -i "ayush.pem" ec2-user@43.205.124.103

      ##_
     _###_
    _####_
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  _####_
 _####_
#/_#####
V~> https://aws.amazon.com/linux/amazon-linux-2023

Last login: Wed Jul  8 12:10:41 2026 from 152.59.87.147
[ec2-user@ip-10-0-1-209 ~]$ sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-1-209 ~]$ curl https://www.google.com
<!doctype html><html itemscope="" itemtype="http://schema.org/WebPage" lang="en-IN"><head><meta content="text/html; char
set=UTF-8" http-equiv="Content-Type"><meta content="/images/branding/googleg/1x/googleg_standard_color_128dp.png" itempr
op="image"><title=Google></title><script nonce="GsNB23ZJw3q-ZNV2iEHSOA">(function){var _g={kEI:'9UJOapbokFGT8L0P1lm26Aw'
,kEXPI:'0,18168,1286035,2935842,48791,30022,6397,9708,138466,206330,301150,5231650,642,97,26,62191862,65173,138341,37805
```

This architectural deployment successfully demonstrates how to maintain a highly secure network layout on AWS. By leveraging a **NAT Gateway**, backend servers (like databases or app layers) are isolated from unauthorized inbound public access while retaining the one-way capability to safely fetch system dependencies, run patches, and download necessary code deployments.